IMPLEMENTATION PLAN · V1.0 · 27 OCT 2026

SnapClaim

Point a phone at damage, speak one line — get a claim-ready report where every number is measured, cited, and checked by an independent model before it ships.

Event	Meta Global AI Developer Hackathon · Oct 26 – Nov 4, 2026
Track	Indie (solo) · Theme: Build for Your World
Builder	Parithosh Varma · from India
Repo	github.com/Parithosh-Varma/snapclaim (public)
Deadline	Nov 4, 9:00 PM ET = **Nov 5, 7:30 AM IST** — submit Nov 4 evening IST

Thesis. A claim is an argument — and arguments need evidence, not a confident-sounding model. SnapClaim is a measuring instrument for the damage claims that never see a surveyor: photo + one spoken line in, an evidence-chained, replayable report out. **Computed, not generated.**

1 Executive summary

Households, tenants, small shops and two-wheeler owners routinely suffer damage that is **economically too small for a professional surveyor** but too contested to settle with a phone photo. The result: deposit forfeitures, undervalued claims, and "trust me" arguments. SnapClaim turns a phone into a documented measurement session:

- **SAM 3.1** segments the damage at pixel level (crack / stain / peel / dent).
- **Deterministic math** — not an LLM — converts masks into percentages, calibrated lengths and severity bands.
- **Muse Spark 1.3** writes the annexure, where every sentence must cite a measured number or the build fails (`EvidenceError`).
- **Spark Open** (Meta open weights, via Baseten) re-measures independently; anything it cannot reproduce is printed as **DISPUTED**, never hidden.
- All visuals are **code-drawn SVG** — the report renders deterministically, identically, on any machine.

This is the receipts doctrine applied to physical damage: *measured vs guessed*, with the guessing machine called out, corrected, and kept on the record.

2 Why it can win

SIGNAL	EVIDENCE	WHY IT MATTERS
Measured, not generated	Every published number is computed from pixels and replayable from a receipts file	A judge can verify any report live — the credibility is the demo
Felt problem	Deposit disputes, small claims, monsoon damage — too small for a surveyor, too contested for a photo	Immediate, human stakes the audience recognizes
Meta tooling depth	Four Meta models with visibly distinct jobs; SAM 3.1 is the flagship "wow" model	The tooling pillar is designed in, not bolted on
Demo quality	Masks bloom → annexure writes itself → one finding flips **DISPUTED** in red → correction → co-signed	A three-minute arc with visible conflict and resolution
Bonus eligibility	Open-weights skeptic (Spark Open), agentic workflow, live replay	Several bonus shots, not just the main track

3 The loop

```
📷 damage photo (+ coin / ruler for scale)  +  🎙 one spoken line
   │
   ▼
1 · SAM 3.1 — pixel masks: "crack", "stain", "peel", "dent"
   │
   ▼
2 · DETERMINISTIC MATH — no LLM in the measurement
      damaged ÷ component area → % damaged
      crack length via mask skeleton × px→mm scale from reference object
      no scale present → percentages only, labeled as such
   │
   ▼
3 · Muse Spark 1.3 — annexure text; every sentence cites a measured number
      EvidenceError if a claim has no citation → build fails
   │
   ▼
4 · Spark Open (Baseten, open weights) — independent skeptic re-measures
      irreproducible findings → printed DISPUTED in red
   │
   ▼
5 · Code-drawn SVG / PDF — masks over photo, receipts table, timestamp, replayable
```

> **Why "computed, not generated" matters:** the judge can replay any report from photo + calibration metadata + receipts file and get the identical numbers. That property — not model eloquence — is the product.

4 Architecture & repository layout

Components

- **web/** — PWA: camera capture, scale-reference guide, voice capture, report view
- **proxy/** — Cloudflare Worker: SAM/Spark/Transcribe calls; keys never on device
- **engine/** — pure TS, zero deps: mask→math, calibration, severity, evidence model
- **findings/** — Spark findings + EvidenceError gate + skeptic merge
- **render/** — deterministic SVG/PDF annexure
- **receipts/** — JSONL evidence chain + replay CLI
- **bench/** — fixtures, held-out set, baselines, metrics page

Repository tree (target)

```
snapclaim/
  src/{web,proxy,engine,findings,render,receipts,bench}/
  fixtures/{damage,scale-ref,receipts}/
  docs/{design.md, plan.pdf, eval-report.md}
  tests/
  .env.example  ·  package.json
  DEV-LOG.md  ·  SUBMISSION.md
  README.md  ·  LICENSE
```

Same TypeScript code runs in the Worker (production/demo) and in Node (dev/bench) — one codebase, two runtimes, zero framework churn.

5 Meta model wiring & spend discipline

PURPOSE	ENDPOINT	MODEL	GUARD
Damage masks	`https://api.meta.ai/v1`	`sam-3.1`	Mock-first; per-call estimate
Evidence-gated annexure	`https://api.meta.ai/v1/chat/completions`	`muse-spark-1.3`	Spend guard
Independent skeptic (open weights)	`https://inference.baseten.co/v1`	`meta-models/muse-spark-open-1026`	Spend guard
Voice intake	`https://api.meta.ai/v1`	`muse-voice-transcribe-1.0`	Mock audio fixtures

- **Mock mode is the default in CI and dev:** recorded fixtures, zero API calls, zero spend.
- `SPEND_GUARD_USD` caps live spend; every live call logs estimated + actual cost into the receipts stream.
- Hindi / Telugu / Hinglish voice notes are a first-class demo path — Muse Voice Transcribe handles code-switching.
- No image generation is used anywhere (Muse Image is geo-blocked from India): every visual is drawn in code from masks + math.

6 Deterministic measurement spec

6.1 Geometry

- `areaRatio = damageMaskPx / componentMaskPx` — always computed, always printable.
- `crackLengthMm = skeletonPx × scaleMmPerPx` where `scaleMmPerPx` derives from the reference object's known dimension in the same image plane.
- Reference present but occluded/low-confidence → fall back to percentages; the fallback itself is a printed receipt.

6.2 Severity (configurable, transparent)

BAND	RULE (DEFAULT THRESHOLDS, PRINTED IN EVERY REPORT)
Cosmetic	< 1% area, no structural line detected
Moderate	1–5% area, or crack length < 150 mm
Significant	> 5% area, or crack ≥ 150 mm, or ≥ 3 separate regions

Thresholds are defaults, versioned in config and echoed in the report; they are not regulatory standards and the report says so.

6.3 Evidence model (receipts)

```
{
  "claim": "crack runs 182 mm across the north wall panel",
  "value": 182, "unit": "mm",
  "derivation": { "mask": "crack#2", "skeletonPx": 913,
                  "scale": { "ref": "coin-10inr", "px": 287, "knownMm": 27 } },
  "skeptic": "CO-SIGNED" | "DISPUTED: scale marker occluded (confidence 0.41)",
  "status": "signed" // cannot publish "signed" without both models agreeing
}
```

7 Evaluation protocol

- **Fixtures:** real damage photos with and without scale references; public damage-image datasets for volume testing.
- **Protocol:** the reasoning model receives only measured facts and named evidence — never the raw photo alone. Otherwise we would be measuring vision comprehension, not the measurement pipeline.
- **Baselines:** (a) model-only qualitative estimate ("looks moderately damaged"), (b) photo report with no measurements. **Headline metric = the guessed-vs-measured gap.**
- **Reproducibility:** held-out split published before tuning; one-command replay of any report from receipts; weak rows printed, not buried.
- **Mock-mode CI:** the whole eval suite runs offline against recorded responses — zero API spend; one live run archived as a JSON artifact.

8 Demo video (3 minutes)

TIME	BEAT
0:00	A real crack on a real wall, phone in hand: "The surveyor comes next month. The deposit is due Friday."
0:20	Snap → masks bloom → speak one Hinglish line → annexure writes itself with green receipts
1:10	One finding flips **red**: skeptic disputes length (scale marker occluded) → human corrects → green, co-signed
2:20	Receipts page: percentages, evidence table, held-out numbers vs baselines; `snapclaim replay` reconstructs the chain
2:50	Close: "Computed, not generated — because a claim is an argument, and arguments need evidence."

9 Build schedule (solo, part-time)

DAY	DATE	PHASE	EXIT CRITERIA
1	Oct 27	Admin + repo + fixtures	Credits / Baseten / Modal forms submitted; repo public; 3–5 real photos; DEV-LOG started
2–4	Oct 28–30	Deterministic core	Mask→math→receipts green in mock mode; EvidenceError gate tested; zero API spend
4–5	Oct 30–31	Live model path	Spark findings + Spark Open skeptic behind spend guard; DISPUTED path demoed
5–6	Oct 31–Nov 1	Renderer + PWA	SVG/PDF annexure; camera + voice capture to report end-to-end
6–7	Nov 1–2	Evaluation	Held-out runs; metrics page; weak rows printed
7–8	Nov 2–3	Polish + video	3-min video recorded; submission fields drafted
9	Nov 4	Submit	Form + public repo + video live before evening IST

10 Risk register

RISK	MITIGATION
No real damage available to photograph	Public damage datasets for pipeline dev; hunt one authentic case for the final video; never fabricate numbers for the money shot
Single-photo scale drift / perspective error	Percentages primary; absolute mm only with confident reference; occlusion → labeled fallback
SAM mask quality on textured walls	Multi-prompt masks + component classifier; degrade gracefully to region estimates with confidence labels
Spark "confidently cites" wrong number	EvidenceError gate: numbers must match receipts values exactly; skeptic re-measure; DISPUTED path printed
Model API hangs / rate limits	Timeout + retry + racing (2 requests, first-stream-wins); mock mode for demos if needed
Baseten key delayed	Open-weight bonus leg waits; Spark 1.3 path unaffected; skeptic mocked until key arrives
Insurance-advice adjacency	Documentation-only positioning; no coverage/legal decisions; human decides
Scope creep (accounts, filing, advice)	Scope cuts enforced (§11); anything new goes to IDEAS-PARK.md until December

11 Scope cuts

No advice engine · no claim filing integrations · no accounts or backend state · no mobile-native app · no image generation · no more than two baselines. Ship the loop, prove the numbers.

12 Compliance guardrails

- Non-medical, non-advisory; no coverage or legal decisions; no medical imagery.
- No geo-circumvention; Muse Image is never used (India geo-block); no VPN.
- Credits claimed only via the builder's own registration email; no pooling.
- Mock-mode CI proves reproducibility; every claim replayable from receipts.
- Build-time tooling disclosed in DEV-LOG.md (the submission form asks for it).

13 Submission checklist

ITEM	DETAIL
Team name / Track	Solo team · Indie track · Problem-statement radio per form
Public repo	✔ github.com/Parithosh-Varma/snapclaim
Video	3-minute YouTube/Vimeo/Véo link, recorded Nov 2–3
Meta tooling field	Receipts from DEV-LOG: SAM 3.1 / Spark 1.3 / Voice Transcribe / Spark Open
Credits claim	$150 by Nov 4 9 PM ET — exact registration email + 16-digit team_id; Visa/MC/AmEx (no RuPay); ~₹1,198–1,247 hold releases
Baseten	Key by email; use only meta-models/muse-spark-open-1026
X post (viral bonus)	Clip: masks bloom + "computed, not generated"
Consent checkbox	Grants marketing rights — watermark the video

14 v0 acceptance gate

> **Gate:** before any dashboard/renderer polish — on the held-out fixture set, measured percentages must beat the model-only qualitative baseline on the guessed-vs-measured gap, every published claim must carry a receipts entry, skeptic disputes must print as DISPUTED, and the full report must replay identically from the receipts file. All of it green in mock mode, zero API spend.

If the gate fails: simplify to percentages-only measurement (no absolute units), keep the receipts discipline, never fake a number.